

NFS Server

Network File System · Linux System & Network Administration

1. Service Fundamentals

NFS (Network File System) allows a client to mount a remote directory and use it as if it were local storage. It is the traditional Unix/Linux network filesystem protocol, ideal for shared home directories and centralized storage between Linux hosts.

Item	Details
Server Package	nfs-kernel-server
Client Package	nfs-common
Port	TCP/UDP 2049 (NFS) · 111 (portmapper for v3)
Export File	/etc/exports
Service Unit	nfs-server.service

2. Architecture

The NFS server exports directories listed in /etc/exports. Clients mount those exports with mount -t nfs. NFSv4 uses a single port (2049) and is simpler than NFSv3, which relies on multiple RPC services.

3. Installation

On the server:

Command:

```
sudo apt update
```

Command:

```
sudo apt install nfs-kernel-server
```

On the client:

Command:

```
sudo apt install nfs-common
```

Enable and start the server service:

Command:

```
sudo systemctl enable --now nfs-server
```

4. Configuration

Create the directory to export:

Command:

```
sudo mkdir -p /srv/nfs/shared
```

Command:

```
sudo chown nobody:nogroup /srv/nfs/shared
```

Command:

```
sudo chmod 755 /srv/nfs/shared
```

Edit the exports file:

Command:

```
sudo nano /etc/exports
```

Add this line (adjust the subnet as needed):

```
/srv/nfs/shared 192.168.100.0/24(rw, sync, no_subtree_check, no_root_squash)
```

Apply the exports and restart:

Command:

```
sudo exportfs -ra
```

Command:

```
sudo exportfs -v
```

Command:

```
sudo systemctl restart nfs-server
```

On the client — mount the share:

Command:

```
sudo mkdir -p /mnt/nfs
```

Command:

```
sudo mount -t nfs 192.168.100.10:/srv/nfs/shared /mnt/nfs
```

For a permanent mount, add this line to /etc/fstab:
192.168.100.10:/srv/nfs/shared /mnt/nfs nfs defaults,_netdev 0 0

5. Important Export Options

Option	Meaning
rw / ro	Read-write or read-only
sync	Write to disk before acknowledging (safer)
async	Faster but risk of data loss on crash
no_subtree_check	Disable subtree checking (recommended)
root_squash	Map remote root to anonymous (default, safer)
no_root_squash	Allow remote root to keep root privileges

6. Verification

On the server — list active exports:

Command:

```
sudo exportfs -v
```

From a client — query available exports:

Command:

```
showmount -e 192.168.100.10
```

Confirm the mount:

Command:

```
mount | grep nfs
```

7. Troubleshooting

Problem	Check	Solution
access denied by server	/etc/exports · exportfs -v	Client IP must match export; re-export
Connection timed out	firewall, nfs-server status	Open port 2049; start the service
Permission denied after mount	ownership, root_squash	Adjust ownership or options carefully
Stale file handle	server reboot / path change	umount -f and remount

8. Security

- Export only to required subnets (never * in production).
- Prefer root_squash (the default).
- Firewall: allow 2049 only from trusted networks.
- For strong authentication consider NFSv4 with Kerberos (sec=krb5).

9. Real-World Scenario

A development team needs a common /data volume. The administrator exports /srv/nfs/data to the development subnet with rw,sync,root_squash. Clients mount it via fstab with the _netdev option so the mount waits for the network.

10. Essential Commands

Command	Purpose
sudo apt install nfs-kernel-server	Install NFS server
sudo exportfs -ra	Apply export changes
sudo exportfs -v	List active exports
showmount -e SERVER	Query remote exports
sudo mount -t nfs ...	Mount on client

Command	Purpose
<code>sudo systemctl status nfs-server</code>	Service status

11. Important Points (Interview / Viva)

1. NFS lets clients mount remote directories as local filesystems.
2. Main port: TCP 2049 · Config: /etc/exports.
3. `exportfs -ra` reapplies exports after changes.
4. `root_squash` maps remote root to anonymous (safer).
5. NFSv4 is simpler (single port) than NFSv3.